

MODEL QUESTION PAPER

TED (15) – 5211

Reg. No.

(REVISION -2015)

Signature

DIPLOMA IN ENGINEERING/TECHNOLOGY

INDUSTRIAL INSTRUMENTS - II

[Time : 3 hours]

(Maximum marks : 100)

PART – A

(Maximum marks : 10)

Marks

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Define Reynolds number.
2. Write the characteristics of turbulent flow
3. State coriolis principle
4. Define dewpoint
5. Write the principle of operation of strain gauge (5x2 = 10)

PART – B

(Maximum marks : 30)

II Answer any five of the following questions. Each questions carries 6 marks.

1. Explain the principle of operation of variable area flow meter
2. Describe different types of orifice plate and its tapings
3. Explain working principle of Ultrasonic flow meters
4. Derive an expression for the discharge through a V-notch
5. With a diagram explain magnetic drag type tachometer
6. Explain the working of LVDT accelerometer
7. Describe the method of measurement of speed by non-contact type instruments

(5x6 = 30)

PART – C

(Maximum marks : 60)

(Answer one full questions from each unit. Each full question carries 15 marks)

UNIT - I

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| III | (a) Explain the construction and working of Rotameter | 8 |
| | (b) Compare the properties of laminar and turbulent flow | 7 |

OR

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| IV | (a) Describe the construction & working of orifice plate flow meter | 8 |
| | (b) Explain working of Dahl tube | 7 |

UNIT – II

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| V | (a) Explain the working principle of Electromagnetic flow meters | 8 |
| | (b) Describe Construction of Hot wire anemometer | 7 |

OR

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| VI | (a) Describe Construction & working of Reciprocating piston type flowmeter | 8 |
| | (b) Explain methods of flow measurement by open channel flow meters | 7 |

UNIT – III

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| VII | (a) Explain the principle and working of Say bolt's viscometer | 8 |
| | (b) Describe the principle and working of hydrometer | 7 |

OR

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| VIII | (a) Explain the operation of wet and dry bulb psychrometer | 7 |
| | (b) Define Humidity, relative humidity, absolute humidity and dew point | 8 |

UNIT –IV

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| IX | (a) Explain the construction & operation of Drag – cup tachometer | 8 |
| | (b) Explain the working of absorption Dynamometer | 7 |

OR

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| X | (a) Explain the construction & working of piezoelectric accelerometer | 8 |
| | (b) Explain the working of transmission Dynamometer | 7 |